

Hybrid Tables



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Section Introduction-Hybrid Tables Topics

- What are Hybrid tables.
- LAB-Create Hybrid tables.
- What is Unistore ?
- Snowflake Architecture and Hybrid tables. (Did Snowflake Architecture change ?).
- Limitation of Hybrid tables.
- Summary.

What is a Hybrid table ?

Hybrid table is a Snowflake table type that is designed for OLTP or transactional application.

Transactional applications need low latency and high throughput which is provided by Hybrid tables.

Hybrid table features

- Primary key (mandatory)
- Unique key constraint.
- Referential Integrity constraint.
- Index

We can create hybrid tables with the syntax

`CREATE HYBRID TABLE <tablename >`

What is Unistore ?(Uni means one)

There are 2 types of workloads that a database can encounter

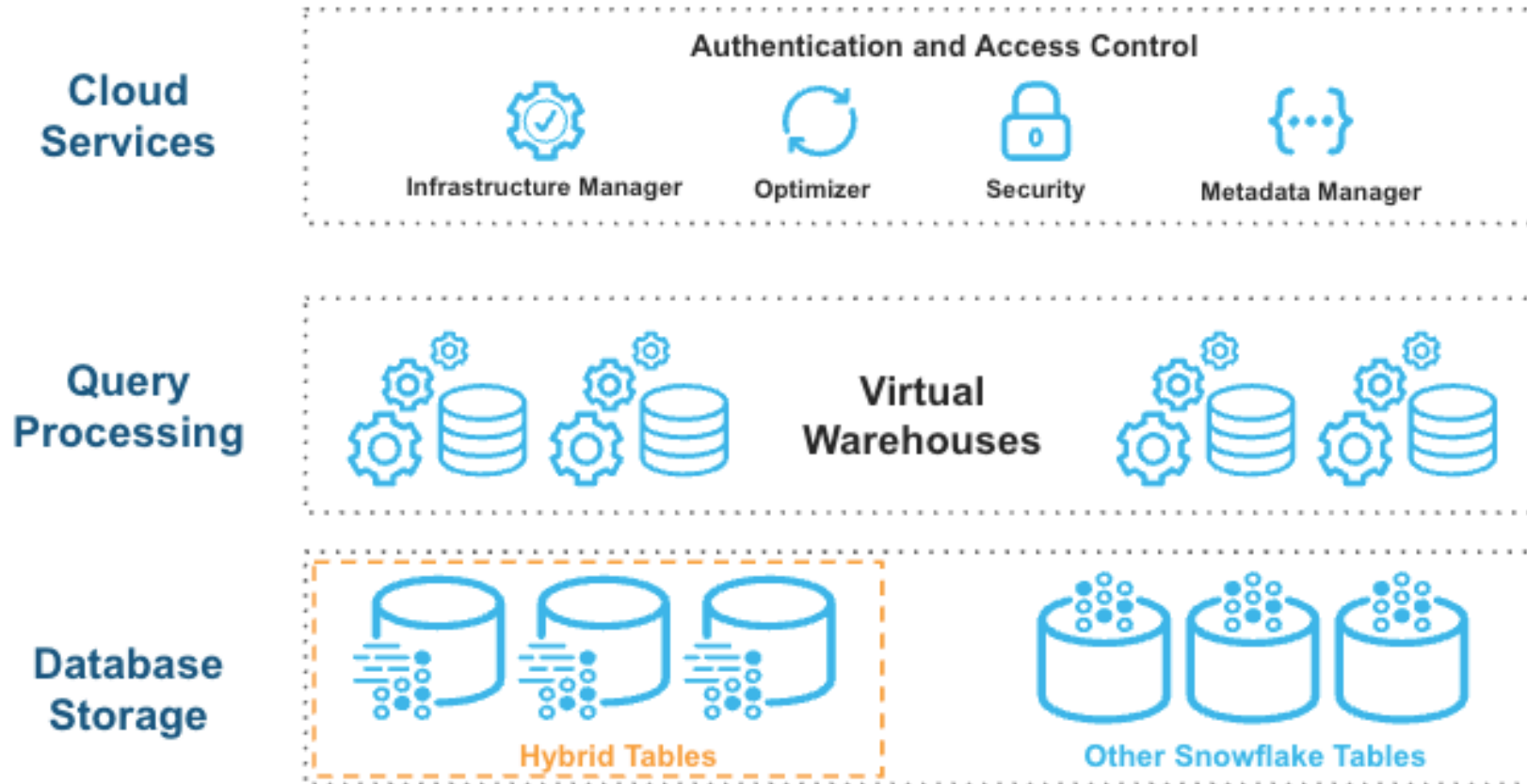
- Analytical workload.
- Transactional workload.

Different databases are used for analytical and transactional workloads as some databases are more suited for transactional workloads and some are more suited for analytical workloads.

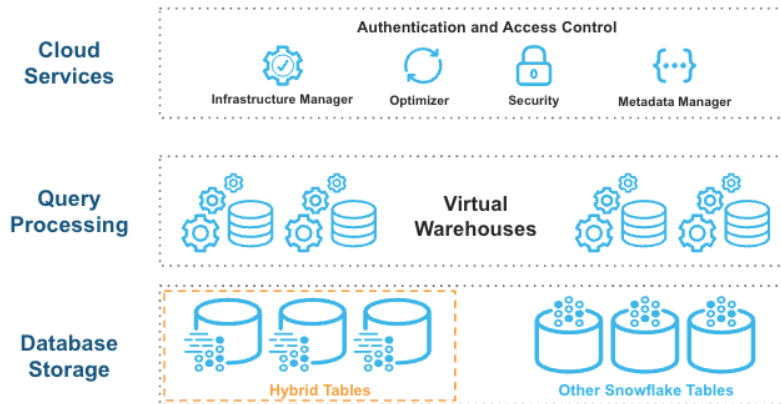
Unistore is Snowflake bold claim that Snowflake cloud database with Hybrid tables is well suited for transactional workloads and Snowflake non-hybrid tables are well suited for analytical workloads making it a single store (uni.) for processing Analytical and transactional workloads

Snowflake hybrid tables and non-hybrid tables can be created in the same database and joined with each other. Since both type of workloads can be handled by Snowflake database, snowflake calls it Unistore. A single store that can handle both analytical and transactional workloads.

Snowflake Architecture and Hybrid tables.



Snowflake Architecture and Hybrid tables.



- No change in the Cloud Services layer.
- No change in the Query processing layer.
- Addition of row store to store hybrid table data in row-oriented manner with secondary columnarization.
- Hybrid tables consume more space than non-hybrid table.
- Hybrid tables consume additional serverless credits.
- Indexes when created consume space.

Limitation of Hybrid table : Hybrid tables do not support

- Clustering Keys
- Time Travel/Data Retention Period/UNDROP
- Cloning
- Fail-safe
- Dynamic Tables
- Materialized Views
- Query Acceleration Service
- Search Optimization Service
- Snowpipe
- Streams
- Data sharing
- Periodic rekeying
- Tri-secret secure encryption
- Collations

Summary

- Hybrid tables are designed to support transactional applications and workloads.
- Hybrid tables must be defined with a PK and the PK is enforced.
- Hybrid tables support unique key and referential integrity constraints.
- Hybrid tables support indexes which help improve select performance.
- Hybrid table data is stored is row-oriented for transactional performance.
- Snowflake architecture has added row store to accommodate hybrid table data.
- Unistore means a single data store that can support analytical and transactional workloads.
- Hybrid table row-oriented data is re-stored in columnar format for analytical workload performance when joining with non-hybrid tables.
- Hybrid tables consume additional storage and serverless credits for table maintenance.
- Hybrid tables and non-hybrid tables can be created in the same database.
- Hybrid tables suffer from several limitations including lack of time travel and fail-safe.
- Hybrid tables are created by using **HYBRID** in the create table statement.

Thank you



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